

DILIPKUMAR NEHRU

Backend Software Development Engineer — Distributed Systems, APIs & Microservices

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PROFESSIONAL SUMMARY

Backend Software Development Engineer with 4+ years designing and operating large-scale, multi-tiered distributed systems and RESTful/microservice APIs (Node.js, Express.js, NestJS, MongoDB, MySQL) processing 50,000+ daily requests at 99.5% uptime for enterprise CRM, payments-adjacent, and analytics platforms. Strong Computer Science fundamentals in data structures, algorithms, and object-oriented design, applied to database performance optimisation, asynchronous/event-driven processing (RabbitMQ), and fault-tolerant service design. Full ownership of the software development life cycle — coding standards, code reviews, source control, build/release processes, and production operations — with a track record of resolving 30+ production incidents and cutting MTTR by ~45%. QA automation background (Selenium, Postman, JMeter) adds end-to-end quality rigor from design through release.

CORE QUALIFICATIONS ALIGNMENT

- 3+ years of non-internship, full software development life cycle experience: coding standards, code reviews, GitFlow source control, build/release processes, testing, and production operations.
- Non-internship design/architecture experience: designed multi-tenant microservice architectures, RabbitMQ-driven event pipelines, and schema/data-consistency standards across services.
- Programming: JavaScript (ES6+) and Node.js as primary general-purpose languages, with working knowledge of Python and Java; comfortable picking up additional languages/stacks (C++/Java fundamentals from CS coursework and OOP design work).
- Computer Science fundamentals: object-oriented design, database/query optimisation, distributed messaging, and reliability/scaling patterns applied in production systems handling 50,000+ requests/day.

CAREER HIGHLIGHTS

- Designed a RabbitMQ-driven, real-time CDR processing pipeline with retry/delivery-log mechanisms — a large-scale, multi-tiered distributed system supporting a cloud telephony SaaS platform.
- Reduced MongoDB query latency by 35% through indexing, query optimisation, and connection pooling on a high-traffic enterprise CRM.
- Cut asynchronous job completion time from 8s to under 2s by redesigning queue consumers and introducing dead-letter queue error handling for an analytics platform.
- Resolved 30+ production incidents, improving system stability and reducing Mean Time to Resolution (MTTR) by ~45%.

WORK EXPERIENCE

Backend Engineer – Software Developer | Triotech Bizserve Private Limited | Bhopal *Jun 2026 – Present*

- Design and develop backend services for Ringingo, a multi-tenant cloud telephony SaaS platform supporting real-time CDR processing, webhook delivery, and third-party API integrations — a large-scale, multi-tiered software system used across multiple tenant organizations.
- Architected a RabbitMQ-driven real-time CDR pipeline with timezone-aware scheduled webhooks, incremental cursor-based delivery, and retry/delivery-log mechanisms, applying reliability and fault-tolerance design patterns to a distributed messaging system.
- Built a Token-to-API Key management system with JWT-based authentication (x-api-key and Bearer token mechanisms), including key refresh and expiry handling, to secure inter-service and third-party API access.
- Delivered features end-to-end across NestJS/Node.js, MongoDB, and microservices, plus React/Next.js with Tailwind CSS, for secure, scalable multi-tenant applications — owning design through deployment.

Software Engineer – Backend Developer | Zuqo Corporation India Pvt Ltd | Bengaluru *Dec 2023 – May 2025*

- Designed and maintained scalable RESTful APIs (Node.js, Express.js) serving 50,000+ daily requests for enterprise web applications at 99.5% uptime — direct experience building and operating large-scale, distributed backend systems.
- Reduced MongoDB query response times by 35% via strategic indexing, query optimisation, and connection pooling on the NovoNordisk CRM platform.

- Implemented asynchronous background task processing with RabbitMQ, cutting average job completion time from 8s to under 2s for UrbanPiper AI Analytics — an event-driven, multi-tiered processing architecture.
- Standardised schema design patterns across MongoDB and MySQL microservices, reducing inter-service data inconsistencies by 60% — applied object-oriented and data-modeling principles to distributed data ownership.
- Collaborated in a 6-member cross-functional Agile team across 2-week sprints, consistently shipping features on schedule with zero critical production rollbacks.
- Debugged and resolved 30+ production incidents, improving application stability and reducing MTTR by ~45% — engaging multiple teams to design, develop, and support production software.
- Enforced clean code standards, GitFlow branching, and code review practices, reducing post-merge defect rate by 25% — contributing to a full software development life cycle culture of quality.
- Implemented JWT authentication and OAuth 2.0 authorisation across multiple APIs, eliminating session vulnerabilities and enabling secure third-party integrations.
- Integrated Socket.io for real-time WebSocket event notifications and live dashboard updates, reducing client-side polling overhead by 80%.

Software Test Engineer – QA Automation | Zuqo Corporation India Pvt Ltd | Bengaluru *Apr 2022 – Dec 2023*

- Built and maintained a Selenium automation suite covering 200+ functional and regression test cases, cutting manual testing effort by 40% — strengthening operational excellence and release quality.
- Performed API testing and backend validation using Postman and JMeter, identifying 80+ defects before production deployment.
- Created structured bug reports, test plans, and troubleshooting runbooks adopted as the team's QA documentation standard.
- Partnered directly with backend developers to triage and resolve defects, reducing average bug resolution cycle from 5 days to 2 days.

PROJECT HIGHLIGHTS

NovoNordisk CRM — Healthcare Enterprise Platform

- Architected scalable backend APIs (Node.js, Express.js, MongoDB) handling 50,000+ daily requests for a global pharmaceutical CRM — a large-scale, multi-tiered system with strict reliability requirements.
- Implemented JWT authentication, OAuth 2.0, field-level encryption, and RBAC for HIPAA-aligned secure data handling of sensitive records.
- Optimised 15 high-frequency MongoDB aggregation queries, cutting dashboard load times by 35% for 500+ concurrent users.

UrbanPiper AI Analytics — F&B Analytics Dashboard

- Built event-driven backend microservices with RabbitMQ and Socket.io to process and push real-time analytics streams to 100+ restaurant client dashboards — an example of distributed, fault-tolerant system design.
- Reduced async job processing latency from 8s to under 2s by redesigning queue consumers and adding dead-letter queue error handling.
- Delivered REST APIs consumed by React dashboards, enabling real-time KPI monitoring with sub-500ms response times.

Client Support & Backend Maintenance

- Provided Tier-2 backend production support for 5+ client applications, diagnosing and resolving scalability bottlenecks under SLA constraints.
- Refactored legacy API endpoints, reducing average response time by 20% and eliminating 3 recurring timeout errors during peak-traffic scenarios.

TECHNICAL SKILLS

Languages	JavaScript (ES6+), Node.js, TypeScript (basic), SQL, Python, Java
Frameworks	NestJS, Express.js, Socket.io, React.js, Next.js
Databases	MongoDB, MySQL, PostgreSQL
Distributed Systems & Messaging	RabbitMQ, Event-Driven Architecture, WebSockets, CDR Processing, Retry/Reliability Patterns

Auth & Security	JWT, OAuth 2.0, RBAC, bcrypt, HTTPS/TLS
Backend Fundamentals	RESTful APIs, Microservices, Data Structures & Algorithms, OOP Design, Async Programming, Webhook Integration
Testing	Postman, Selenium WebDriver, JMeter, API & Manual Testing
DevOps / Tools	Git, GitFlow, Bitbucket, GitHub, Linux CLI
Methodologies	Agile, Full SDLC Ownership, CI/CD Awareness

CERTIFICATIONS

- Selenium Automation & Manual Testing — QSpiders, Bangalore (2022)
- MongoDB Basics (M001) — MongoDB University (Recommended next)
- Node.js Application Developer — OpenJS Foundation (Recommended next)

EDUCATION

Bachelor of Engineering – Electronics & Communication Engineering (ECE) | RVS Technical Campus, Coimbatore (Anna University) | 2020

Diploma – Electronics & Communication Engineering (ECE) | Krishnasamy Memorial Polytechnic College, Cuddalore | 2012 – 2015